Hybrid Wheeled Drone
Ardupilot Rover CUAV X7+

Week 1 scope (Ground Unit / Rover)

Objective: A hybrid platform that can fly as a drone and drive on the ground using two wheels, with strict safety gating so the wheels never operate in the air.

Step 1 - Ground Rover (Skid-Steer) Setup Report

Project: Wheeled Drone (Ground + Flight) - Ground Unit First
Controller: CUAV X7+ (ArduRover v4.6.3 / ChibiOS)
Drive Concept: Skid-Steer (Tank) - left side wheels vs right side wheels
ESCs: 2× FLIPSKY FSESC4.20 50A (VESC-based, 12S capable)
Radio / Video: Skydroid T12 kit (Receiver + Camera link)
Tools Used: Mission Planner/QGC + VESC Tool + SITL (planned)

1) Objective (Step-1 Scope)

This first step focuses **only on the ground unit (Rover)**, starting from a clean and safe configuration, then validating control behavior and autonomy features. The goal is to achieve:

1. **Correct tank steering**: forward / backward / left / right with smooth turning (one side faster than the other).
2. **Reliable safety behavior**: wheels do not spin unexpectedly; controlled engagement/disengagement via radio switch logic.
3. **Solid base for autonomy**: RC + GPS navigation (basic hold, waypoint mission), to be extended later.

Note: The "flying + driving" combination is a later step. For now, we establish a **known-good rover baseline** before merging it with the flight system.

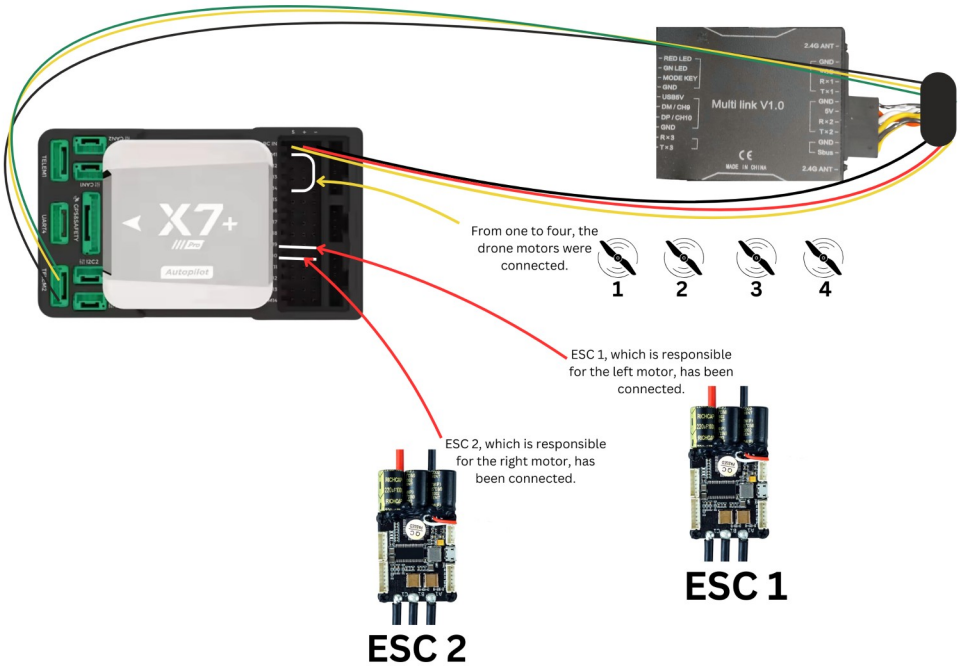
2) Current Understanding of the Hardware Architecture

Mechanical:

- Left side = **2 wheels mechanically linked** (same drive output).
- Right side = **2 wheels mechanically linked** (same drive output).
- Therefore only **two motor channels** are required: **LEFT** and **RIGHT**.

Electrical / Control:

- CUAV X7+ provides RC outputs; two outputs are assigned to drive the two VESC ESCs.
- Each VESC controls its own motor side (Left / Right).
- VESCs are configured using **VESC Tool** (PPM input mapping shown in screenshots).



ArduPilot Rover (Skid-Steer) Key Principle:
Skid-steer rovers are controlled by **two independent throttle outputs**:

- `ThrottleLeft` and **ThrottleRight** mapped to two servo outputs.

3) What We Can Confirm from Your Parameter File + Screenshots
From your provided parameter file (`rover_tank.param`), we can confirm:

- **FRAME_CLASS = 1 (Rover)**
- **FRAME_TYPE = 0**
- **SERVO9_FUNCTION = 73 (ThrottleLeft)**
- **SERVO10_FUNCTION = 74 (ThrottleRight)**
  These functions match ArduPilot Rover skid-steer guidance (ThrottleLeft=73, ThrottleRight=74).

Also observed:

- RC channel options (RC5/RC6/etc.) appear not yet assigned (**`RCx_OPTION = 0`** in the shared file), meaning switch logic still needs to be implemented cleanly for "enable/disable / mode select / safety".

4) Current Problem Statement (As Reported)

1. **When SERVO9/10 are used (ThrottleLeft/ThrottleRight), VESC Tool shows "no signal"** (no valid input reaching the VESC).
2. **When settings are changed to RCIN3/4**, the motors respond but **movement becomes only linear** (forward/backward only), and no effective left/right turning occurs.
3. The desired turning behavior is **simple tank turning**: one side faster/slower than the other, including pivot turns if required.

Interpretation (most likely):

- The VESC input signal path is not consistently coming from the **autopilot RC outputs** (SERVO9/10), OR the VESC is still reading **receiver-side signals** (RCIN) instead of autopilot-generated outputs.
- In other words: the system sometimes behaves like "direct receiver → VESC", which can break skid-steer mixing and safety logic.
- We need a clean separation: **Receiver → Autopilot input**, then **Autopilot output → VESC input**.

5) Step-1 Plan: Clean Rover Setup (from scratch)

This is the exact step-by-step work we will do to build a reliable baseline.

A) Wiring Baseline (must be confirmed during meeting)

Goal: ensure the VESCs receive **PWM/PPM signal from the CUAV outputs**, not directly from the receiver.

- Receiver outputs go to the CUAV input (SBUS/PPM/serial depending on your receiver wiring).
- CUAV **SERVO outputs** go to each VESC PPM input:
- **SERVO9 → Left VESC (Left motor group)**
- **SERVO10 → Right VESC (Right motor group)**

This matches the idea of mapping ArduPilot outputs to functions using the RC output mapping concept.

B) VESC Configuration (each ESC separately, then together)

Using **VESC Tool**:

1. Confirm each VESC is configured for **PPM input** and that the pulse range is correct (center ~1.5ms, min/max around 1.0/2.0ms depending on radio/output).
    - VESC PPM mapping uses pulse start/end/center parameters (shown in your screenshots).
2. Confirm the VESC input monitor changes when CUAV outputs are moved (not when receiver direct moves).
3. Confirm direction (forward/back) and apply inversion on one side if needed (common with mirrored drivetrains).
4. Apply ramping and safe-start parameters to avoid jerks when enabling.

C) ArduRover Clean Configuration (Mission Planner)

1. Load a **clean rover baseline** and then compare/merge only what we need from your existing file (`rover_tank.param`).
2. Set skid-steer output mapping based on official guidance: ThrottleLeft and ThrottleRight must be assigned to two outputs.
    - In your current config that's already: **`SERVO9_FUNCTION=73 and SERVO10_FUNCTION=74.`**
3. Verify servo output ranges: **`SERVOx_MIN / TRIM / MAX`** match VESC expectations (usually 1100/1500/1900 or 1000/1500/2000 depending on calibration).
4. Confirm **RC calibration** and joystick mixing strategy:
    - One stick controls throttle (forward/back)

- One stick controls steering (left/right)
- The rover mixer must convert this into left/right throttle commands.

D) Safety Logic (Ground-unit safety first)

In rover phase, we implement safety as:

- Wheel outputs only active when:
    1. Vehicle is in an allowed mode, and
    2. A dedicated **enable switch** is ON (example CH5), and
    3. Arming state + failsafe behavior is correct (no unexpected output).

Later, when we merge flight + ground, we'll add "in-air inhibit" (landed detection / flight mode gating). For now we establish ground safety correctly.

E) Validation Checklist (bench → floor tests)

1. Bench test with wheels off ground: verify left/right commands are independent.
2. Floor test at low speed: confirm smooth turn (inside wheel slower, outside wheel faster).
3. Confirm failsafe: signal loss stops outputs.
4. Confirm switch-enable behavior: disable switch forces outputs neutral.

6) SITL + Gazebo Verification (before final tuning on hardware)

To reduce trial-and-error and validate the logic, I will reproduce the rover as a **skid-steer model in simulation** and validate mixing + control behavior:

- Use ArduPilot SITL with Gazebo as the external simulator.
- Validate:
    - RC input mapping
    - Steering/throttle mixing
    - Waypoint navigation basics
    - Output behavior and safety conditions (arming/switch logic)

This gives us a "known good" reference before applying final parameters to the real vehicle.

7) Immediate Observations from Logs / Screenshots (What I'd Like to Verify Live)

Based on what you sent, the main red flag is:

- **VESC sees no signal when SERVO9/10 are enabled**, which suggests either:
    1. the VESC PPM signal wire is not connected to the correct CUAV output pins, or
    2. the VESC is still wired to receiver output (RCIN path), or
    3. output signal type / ground reference is wrong, or
    4. the output is not physically "active" on that port group as expected.

This is exactly what we will confirm quickly during the first meeting using Mission Planner servo output test + VESC input monitor.

8) Deliverables for Step-1 (Week-1 Work Product)

At the end of the first week (ground-unit only), you will receive:

- Clean parameter set (exported .param) for Rover skid-steer baseline
- Wiring verification checklist + diagram corrections (if any)
- A short test procedure (bench + ground)
- SITL/Gazebo validation notes (what was validated and how it matches real behavior)

9) Next Step: Short Meeting (to confirm wiring + goal behavior)